

ShopSleek

E-commerce Platform

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Project Overview

- Online shopping platform for various products
- User registration and account management
- Guest shopping capabilities
- Product search and filtering
- Shopping cart management
- Secure checkout process
- Order tracking
- Admin dashboard for system management
- Multi-vendor support system

Key Features

- Intuitive User Interface - Clean, responsive design optimized for all devices
- Advanced Search & Filtering - Find products quickly with dynamic filtering
- Secure Payment Processing - PCI-DSS compliant payment gateway integration
- Order Management - Track orders from placement to delivery
- User Accounts - Personal profiles, wishlists, and order history
- Admin Dashboard - Complete control over products, orders, and users
- Analytics & Reporting - Sales reports, user activity, and inventory levels

System Architecture

Three-Tier Architecture

- Presentation Layer: User Interface, Admin Dashboard, Vendor Panel
- Application Layer: User Management, Product Management, Order Processing, Payment Gateway
- Data Layer: MySQL Database, File Storage

Technology Stack

- Frontend: React.js, Tailwind CSS
- Backend: Node.js, Express.js
- Database: MySQL
- Server: Node.js
- Hosting: HostingRaja

Test Cases - User Authentication

ID	Test Scenario	Expected Result	Status
TC01	User Registration - Valid Data	Account created with confirmation	Pass
TC02	User Registration - Empty Form	Error message displayed	Pass
TC03	User Registration - Invalid Email	Validation error shown	Pass
TC04	User Login - Valid Credentials	Logged in and redirected	Pass
TC05	User Login - Invalid Credentials	Error message displayed	Pass
TC06	User Login - Blank Fields	Form validation errors	Pass

Test Cases - Product Browsing

ID	Test Scenario	Expected Result	Status
TC07	Product Search - Valid Query	Relevant products displayed	Pass
TC08	Product Search - No Results	Empty results message shown	Pass
TC09	Product Filtering - By Category	Filtered products displayed	Pass
TC10	Product Filtering - Multiple Filters	Correctly filtered results	Pass
TC11	Add to Cart - In Stock Item	Product added to cart	Pass
TC12	Add to Cart - Out of Stock	Error message displayed	Fail

Test Cases - Shopping Cart & Checkout

ID	Test Scenario	Expected Result	Status
TC13	Cart Modification - Update Quantity	Cart updated correctly	Pass
TC14	Cart Modification - Remove Item	Item removed from cart	Pass
TC15	Checkout - Valid Payment Info	Order placed successfully	Pass
TC16	Checkout - Invalid Card Number	Error message displayed	Pass
TC17	Checkout - Invalid CVV	Error message displayed	Pass
TC18	Order Tracking - Valid Order ID	Correct order status shown	Fail

Test Results Summary

Category	Total Tests	Passed	Failed	Success Rate
User Authentication	6	6	0	100%
Product Browsing	6	5	1	83.3%
Shopping & Checkout	6	5	1	83.3%
Admin Functions	8	7	1	87.5%
API Endpoints	10	9	1	90%
Overall System	36	32	4	88.9%

Security Aspects

Data Protection

- Password encryption using strong hashing algorithms
- Sensitive data encryption in database
- Role-based access control mechanisms
- Session management and timeout controls
- Regular security audits and updates

Security Implementation

- SQL Injection: Prepared statements and parameterized queries
- XSS Attacks: Output encoding and content security policy
- CSRF Attacks: Unique tokens for each session
- Session Hijacking: Secure cookies and session regeneration
- Brute Force: Account lockout and CAPTCHA integration

Deployment Architecture

- Hosting: Deployed on HostingRaja with 99.9% uptime guarantee
- Server Configuration: Node.js server with Express.js framework
- Database: MySQL 8.0 with optimized query performance
- SSL Implementation: HTTPS encryption for secure data transmission
- Scalability: Load balancing for high traffic management
- Backup System: Daily automated backups with point-in-time recovery
- Monitoring: Real-time performance and security monitoring
- CDN Integration: Content delivery network for optimized asset delivery

Future Scope

- Mobile Applications - Native Android and iOS apps
- AI-powered Recommendations - Personalized product suggestions
- Advanced Analytics - Customer behavior and sales forecasting
- Voice Commerce - Integration with voice assistants
- Augmented Reality - Virtual product try-on experiences
- Internationalization - Multi-language and multi-currency support
- Marketplace Expansion - Platform for multiple vendors
- Integration with Social Commerce - Buy directly from social platforms

Thank You!

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